```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
#define OLED_RESET -1
#define OLED_ADDRESS 0x3C

const int MIC_PIN = A0;

Adafruit_SSD1306 display(
 SCREEN_WIDTH, SCREEN_HEIGHT,
 &Wire, OLED_RESET
);

int measureSoundLevel() {
 int signalMin = 1023;
 int signalMax = 0;

 unsigned long start = millis();

 while (millis() - start < 50) {
   int sample = analogRead(MIC_PIN);

   if (sample < signalMin) signalMin = sample;
   if (sample > signalMax) signalMax = sample;
 }

 return signalMax - signalMin;
}

void setup() {
 Serial.begin(9600);

 if (!display.begin(
       SSD1306_SWITCHCAPVCC, OLED_ADDRESS)) {
   while (true) {
     // Check OLED wiring and address.
   }
 }
```

```
 display.clearDisplay();
 display.setTextColor(SSD1306_WHITE);
 display.display();
}

void loop() {
 int level = measureSoundLevel();

 Serial.println(level);

 display.clearDisplay();

 display.setTextSize(1);
 display.setCursor(0, 0);
 display.println("SOUND LEVEL");

 display.setTextSize(2);
 display.setCursor(0, 16);
 display.print(level);

 display.drawRect(0, 44, 120, 12, SSD1306_WHITE);

 // Display a relative level, not calibrated decibels.
 int barWidth = constrain(map(level, 0, 100, 0, 116),
                          0, 116);

 display.fillRect(2, 46, barWidth, 8, SSD1306_WHITE);

 display.setTextSize(1);
 display.setCursor(0, 58);
 display.println("Relative level");

 display.display();
}
```